📄 WEEK 3 PROGRESS REPORT – MINOR PROJECT

Project Title

AI-Powered Student Assistance Chatbot for Department of Technical Education, Government of Rajasthan

Group ID

G-1

Branch & Section

AI & DS – B

Date Range

02/02/2026 to 07/02/2026

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1. EXECUTIVE SUMMARY

Current Sprint

Sprint 3

Sprint Goal

Static Analysis and Domain Modeling

Status

On Track

Summary of Work Done

During Sprint 3, the team focused on static analysis of the system. Domain entities related to the AI-powered chatbot were identified based on functional requirements gathered in previous sprints. The relationships between entities such as College, Student, Admission, Fees, Scholarship, Placement, and Administrator were analyzed.

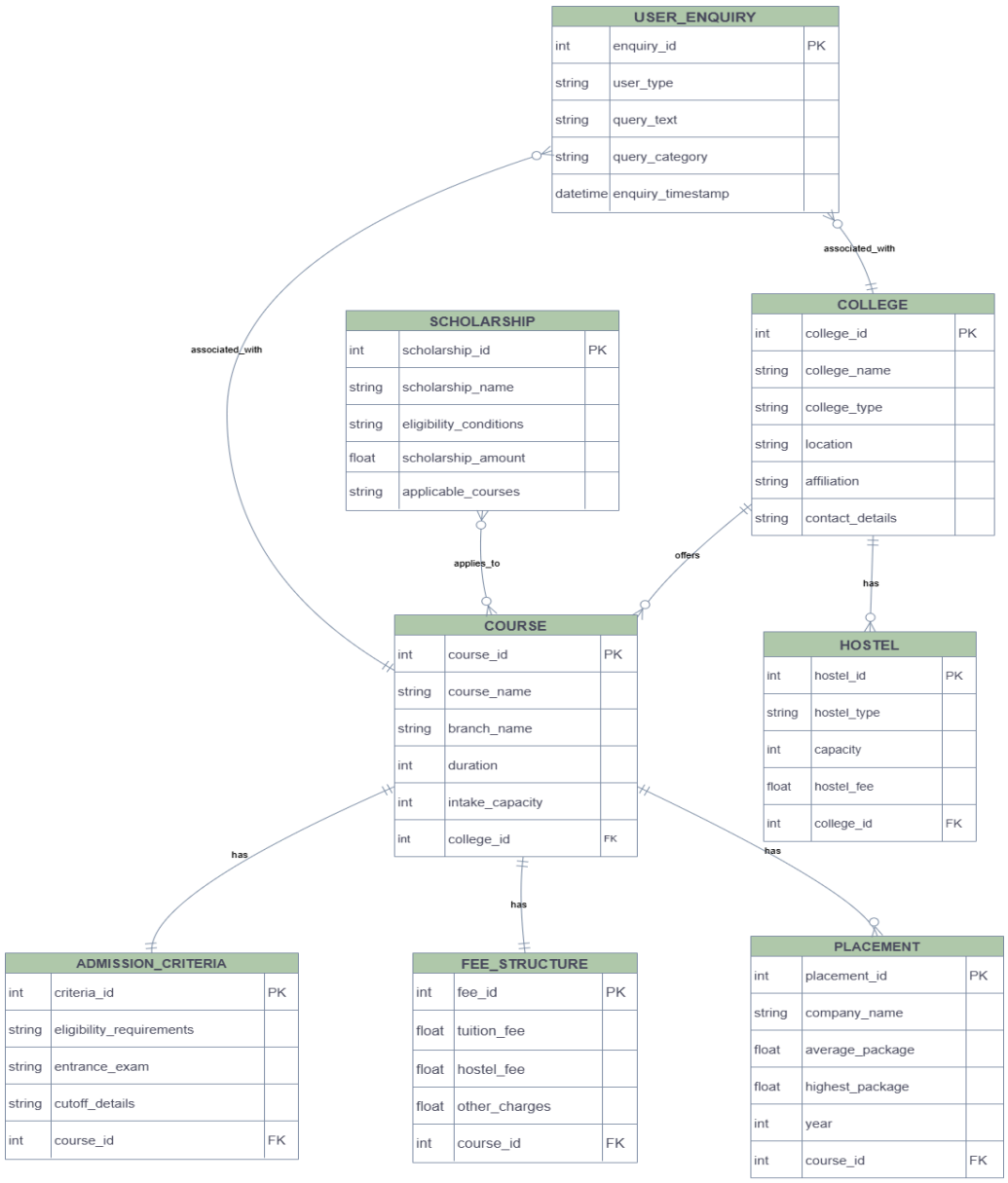
An ER Diagram was prepared to represent a real-time snapshot of system data. Additionally, an Initial Class Diagram (Domain Model) was developed focusing only on attributes and relationships without including methods. This helped in understanding the structural design of the system before implementation.

2. UML & OOAD PROGRESS

✅ UML Diagrams Completed This Sprint

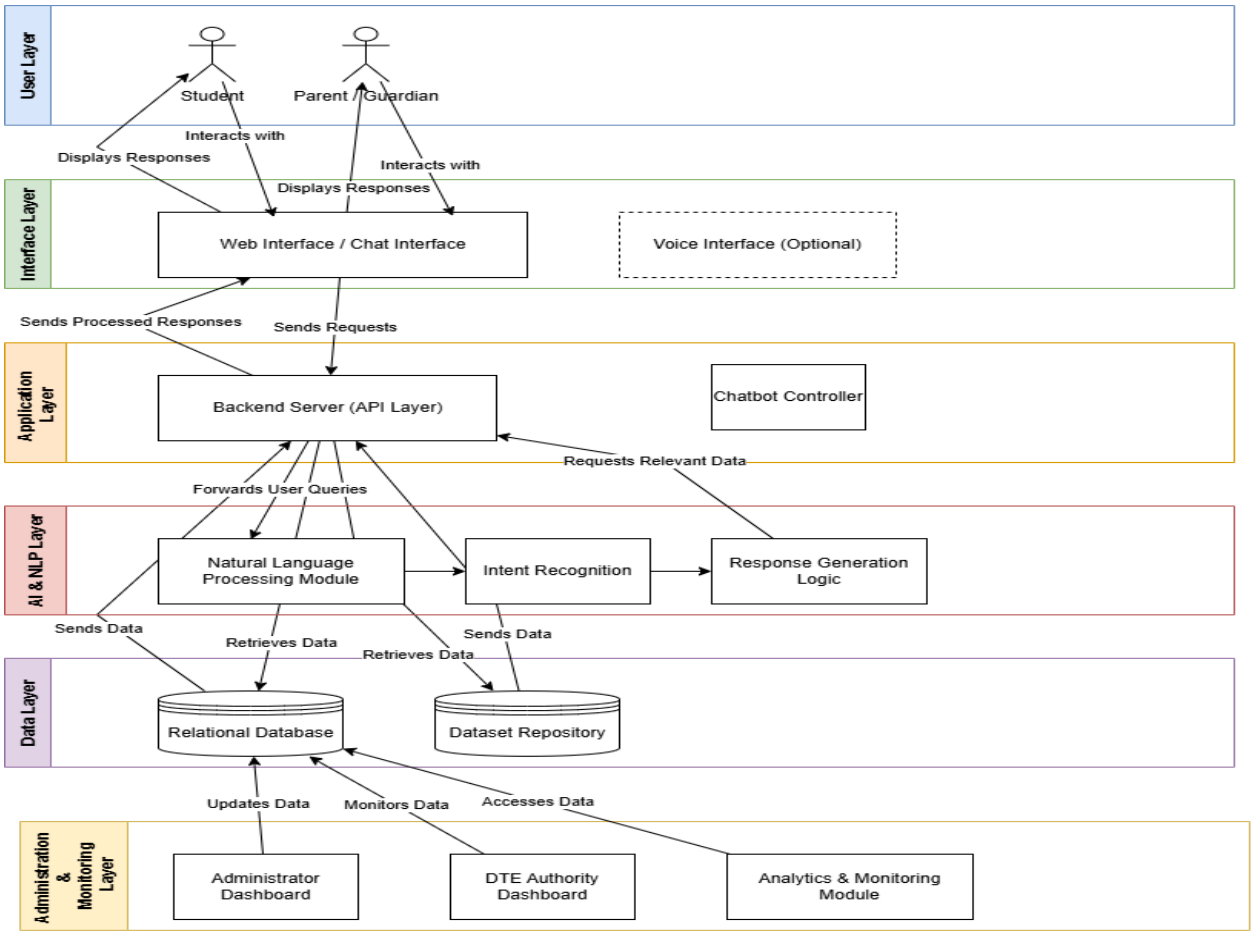
1. ER Diagram

- Modeled real-world instances such as:
  - College: Govt. College
  - Admission: Admission 2026
  - Scholarship: Post Matric Scholarship
- Represented actual data snapshot and relationships between instances.



2. System Architecture

Identified domain entities with attributes only (no methods).



📌 Identified Domain Entities

- Student
- Parent
- College
- Admission
- Fees
- Scholarship
- Placement
- Administrator
- Chatbot

📌 Key Relationships Identified

- College has Admission
- College has Fees
- College offers Scholarship

- College has Placement Records
- Student interacts with Chatbot
- Administrator manages College Data

📌 Object-Oriented Concepts Applied

- Abstraction (identifying core entities)
- Association relationships
- Domain modeling without implementation logic
- Clear separation between structural and behavioral design

3. TECHNICAL EXECUTION (DEVOPS & VERSION CONTROL)

Task Tracking

- Static analysis tasks created in Jira
- Domain modeling tasks assigned to team members

Git Activity

- Versioned UML artifacts
- Separate branch for modeling
- Structured commit history
- Traceability between Jira story and UML deliverable

Active Branches

- main
- uml-modeling

Merged Pull Requests

- Domain model draft merged

CI/CD Status

- Not applicable (design phase)

4. TESTING SUMMARY

Total Unit Tests

- 0 (development phase not started)

Key Issues Identified

- Need for standardizing attributes across all colleges
- Variations in data format from different sources

5. GOALS FOR NEXT SPRINT (SPRINT 4)

- Refine Class Diagram with methods
- Design Database Schema based on domain model
- Begin backend implementation
- Start integrating chatbot query structure

SIGNATURES

Project Guide Signature: _____

Mentor Signature: _____